# Generative AI: Redefining the Future of Software Engineering

**Anita Carleton**, Carnegie Mellon University Software Engineering Institute

**Davide Falessi**, University of Rome Tor Vergata

**Hongyu Zhang**, Chongqing University

**Xin Xia**, Huawei Technologies



*Digital Object Identifier 10.1109/MS.2024.3441889*
*Date of current version: 10 October 2024*

 0740-7459/24©2024IEEE

**THIS SPECIAL ISSUE** about generative AI (GAI) for software engineering refers to the application of generative models and algorithms in software development, testing, maintenance, and evolution.[1] Generative models are machine learning (ML) models that can create new data based on a series of inputs, such as text, image, or code. GAI has the potential to revolutionize the world in which we live, including the way we develop software. Although ML has already provided numerous improvements in many areas of software engineering—such as requirements quality, traceability recovery, design assessment, and testing priority—GAI differs from ML in many ways. ML algorithms are usually used to classify or predict results based on training data, while algorithms generate new data with characteristics similar to training data. GAI techniques are generally more complex and sophisticated than traditional ML techniques since they require modeling the underlying distribution of the data and generating new data that are coherent and meaningful.

In summary, traditional ML has already provided numerous improvements in many areas of software engineering. Still, GAI offers a unique approach to generating new data and content that can further improve software development and maintenance processes. Using GAI algorithms, software engineers can automate many tedious and time-consuming tasks and accelerate the pace of software innovation.

GAI can reasonably support software engineers in any task, such as requirement gathering or design. For example, GAI can help collect requirements by analyzing inputs from stakeholders and generating initial requirements for the software system. GAI can help in the design phase by generating design artifacts, such as wireframes, user interface designs, and architectural diagrams from requirements and standards. GAI is a revolutionary technology, so accurately predicting the software engineering tasks it will be helpful in is challenging.

As industry, government, and academic organizations devise plans for adopting GAI into their business and engineering operations/solutions, there are a couple of options for technology leaders to consider: Companies can leverage GAI in software tools that they use or focus resources on developing GAI solutions on their own. This is a decision that should be made with every organization's unique needs in mind. In the short term, it is much easier, faster, and more cost effective for organizations to leverage trusted tools with GAI capabilities than to develop a GAI solution from scratch. This special issue features five articles where you will see some examples of GAI adoption and a discussion of research directions and challenges as GAI continues to experience transformative growth and application. GAI's applications include automating workflow, designing and creating marketing content, connecting teams to each other, and much more. These articles show that the excitement about this new technology is spreading quickly throughout various industries and departments. It is clear that there is great interest in the field, but we have not yet realized all of its potential. Through applications, such as those in software acquisition and industry, we will continue to better understand and learn its powerful use cases.

## GAI in Acquisition

GAI promises to revolutionize software development and acquisition, much like the advent of cloud computing, Agile, or DevSecOps. It will become an invaluable coding partner for developers, enhancing productivity and reliability.[2] In government and corporate settings, GAI will drive significant changes in how software is developed, maintained, and acquired. Applying GAI to software acquisition, for example, provides the potential for a transformative shift.

To succeed, software acquisition, which refers to obtaining software applications and systems for organizational use, must ensure the software aligns with the acquirers' needs, budget, and technical infrastructure. While the use of GAI to support acquisition tasks is still new, developing sophisticated models is already underway. These models could help with critical tasks such as the following[3]:

- Analyzing complex regulatory policies and standards to identify the most relevant aspects for specific acquisitions and monitoring compliance throughout the system lifecycle.
- Identifying potential risks or threats, including cyberthreats and supply-chain vulnerabilities, to enable proactive security measures and risk mitigation.
- Enhancing data analytics capabilities to forecast project outcomes and risks more accurately.
- Digesting and interpreting large and complex acquisition documents to facilitate decision making.
- Summarizing and updating extensive policy documents, such as directives and memoranda, for greater consistency.

Integrating GAI in acquisition processes presents significant opportunities and challenges. For acquisition professionals, mastering AI-augmented processes is essential to incorporate these tools into existing workflows seamlessly.[4] Training is necessary to manage and collaborate with AI systems effectively, ensuring a smooth transition and enhancing operational efficiency.

It is important to recognize that applying advanced tool support to acquisition tasks—especially GAI-based techniques—is still in its infancy.[5] Further work is needed, therefore, on developing more sophisticated GAI models that can[6]:

- Understand and interpret large and complex acquisition documents.
- Enhance the data analytics capabilities to forecast project outcomes and risks more accurately.
- Create more intuitive interfaces for human–AI interaction to facilitate decision making.
- Conduct comprehensive studies on the long-term impacts and ethics of AI integration into the acquisition process.

GAI can streamline decision making, optimize procurement strategies, and predict future needs more accurately.[7] However, the reliance on AI also necessitates rigorous scrutiny and transparent processes to mitigate biases and ensure the integrity of the data used. This is particularly critical in defense acquisition, where precision and reliability are paramount. While GAI has already demonstrated its potential for significantly enhancing the acquisition process, it is crucial to adopt a balanced approach that combines the strengths of AI with human oversight.

## GAI in Industry

Large language models (LLMs) for code significantly impact the software industry by automating code generation, enhancing productivity, improving code quality, etc. Extensive large code models have been developed, including CodeX, StarCoder, CodeLLaMa, and DeepSeek-Coder. They are varied in the number of parameters and size of the pretraining data. These LLM-based tools could help developers skip tedious preliminary steps, allowing them to dive directly into more engaging and impactful problem-solving tasks. According to a report in *DevPro Journal*,[8] users of CopilotX autogenerate and test 46% of their code, accelerating the coding workflow for DevOps teams by 55%. Beyond expediting development processes, eliminating mundane tasks also enhances developer satisfaction. Close to three-quarters of Copilot users reported reduced frustration and an increased capacity to accomplish more fulfilling work. Inspired by the multiagent techniques, some tools—such as AutoGen,[9] MetaGPT,[10] and AutoAgents[11]—are also built to play different roles in the software development process. Integrating LLMs into software engineering fosters innovation, enabling nonexperts to contribute effectively and easing the programmer shortage. However, the ethical considerations when using LLMs, like bias mitigation and intellectual property integrity are also crucial. LMs pragmatically shape software engineering, democratizing coding, yet careful implementation ensures balanced progress.

## Research Directions

GAI holds immense potential to revolutionize software engineering. However, it also raises several challenges that require more research efforts in the future. For example:

- *Correctness and accountability:* The correctness of the outputs generated by GAI still needs to be further improved. Establishing accountability for GAI-generated outputs is essential. Liu et al.[12] performed an empirical study and found that the code generation performance can be improved by carefully designing prompts to guide ChatGPT. Their results show that enhancing the models' ability to understand the context and user intent could improve the relevance and correctness of the generated outputs.
- *Security and privacy concerns:* GAI in software engineering could introduce new security issues if not properly managed. There are also privacy concerns regarding the ownership and copyright of GAI-generated artifacts. Exploring GAI's security and privacy aspects for SE models has become an important research direction. For example, Wan et al.[13] aim to understand the security and robustness of neural code search models by attacking the models through data poisoning.
- *Collaborative working:* Investigate multiple GAI agents' potential to simulate a collaborative working environment where multiple human and AI engineers can seamlessly coordinate and combine their work.[13] For example, Tao et al.[14] recently proposed an LLM-based multiagent framework for GitHub issue resolution.

### References

1. C. Ebert and P. Louridas, "Generative AI for software practitioners," *IEEE Softw.*, vol. 40, no. 4,

pp. 30–38, Aug. 2023, doi: 10.1109/MS.2023.3265877.
2. A. Carleton et al., *Architecting the Future of Software Engineering: A National Agenda for Software Engineering Research and Development*. Pittsburgh, PA, USA: Carnegie Mellon Univ. Software Engineering Institute, 2021.
3. S. Bellomo, S. Zhang, J. Ivers, J. Cohen, and I. Ozkaya, "Assessing opportunities for LLMs in software engineering and acquisition," Carnegie Mellon University Software Engineering Institute, White Paper, Nov. 2023. [Online]. Available: https://insights.sei.cmu.edu/library/assessing-opportunities-for-llms-in-software-engineering-and-acquisition/
4. J. Robert and D. Schmidt, *10 Benefits and 10 Challenges of Applying Large Language Models to DoD Software Acquisition*. Pittsburgh, PA, USA: Carnegie Mellon Univ. Software Engineering Institute, Jan. 2024.
5. D. Schmidt, A. Carleton, J. Ivers, I. Ozkaya, J. Robert, and S. Zhang, *The Future of Software Engineering and Acquisition With Generative AI*. Pittsburgh, PA, USA: Carnegie Mellon Univ. Software Engineering Institute, Jan. 2024.
6. McKinsey & Company. "What is generative AI?" McKinsey & Company Featured Insights. Accessed 19 Mar. 2024. [Online]. Available: https://www.mckinsey.com/featured-insights/mckinsey-explainers/what-is-generative-ai#/
7. Fan et al., "Large language models for software engineering: Survey and open problems," in *Proc. IEEE/ACM Int. Conf. Softw. Eng., Future Softw. Eng. (ICSE-FoSE)*, Melbourne, Australia, 2023, pp. 31–53, doi: 10.1109/ICSE-FoSE59343.2023.00008.
8. "Codex: Developer replacement or development enhancement?" *DevPro J*. Accessed: Sep. 1, 2024. [Online]. Available: https://www.devprojournal.com/technology-trends/ai/codex-developer-replacement-or-development-enhancement/
9. Q. Wu et al., "AutoGen: Enabling next-gen LLM applications via multi-agent conversation framework," 2023, *arXiv:2308.08155*.
10. S. Hong et al., "MetaGPT: Meta programming for multi-agent collaborative framework," 2023, *arXiv:2308.00352*.
11. G. Chen et al., "AutoAgents: A framework for automatic agent generation," 2023, *arXiv:2309.17288*.
12. Liu et al., "Guiding ChatGPT for better code generation: An empirical study," in *Proc. IEEE Int. Conf. Softw. Anal., Evolution Reeng. (SANER)*, Rovaniemi, Finland, 2024, pp. 102–113, doi: 10.1109/SANER60148.2024.00018.
13. Y. Wan et al., "You see what I want you to see: Poisoning vulnerabilities in neural code search," in *Proc. 30th ACM Joint Eur. Softw. Eng. Conf. Symp. Found. Softw. Eng. (ESEC/FSE)*, 2022, pp. 1233–1245, doi: 10.1145/3540250.3549153.
14. Tao et al., "MAGIS: LLM-based multi-agent framework for GitHub issue resolution," 2024, *arXiv:2403.17927*.

## ABOUT THE AUTHORS



**ANITA CARLETON** leads the Software Solutions Division at Carnegie Mellon University Software Engineering Institute, Pittsburgh, PA 15213 USA. Contact her at adc@sei.cmu.edu.



**DAVIDE FALESSI** is an associate professor at the University of Rome "Tor Vergata," 00133 Rome, Italy. Contact him at d.falessi@gmail.com.



**HONGYU ZHANG** is a professor at the School of Big Data and Software Engineering, Chongqing University, Chonqing 400044, China. Contact him at hongyujohn@gmail.com.



**XIN XIA** is a chief expert of software engineering application technologies at Huawei Technologies, Hangzhou 310000, China. Contact him at in.xia@acm.org.